Supplemental Material

Table S1. Search Strategies for each database.

Database	Search String	Results
SCOPUS	1. "randomized controlled trial " OR "controlled clinical trial" OR randomized OR placebo OR randomly OR trial OR rct 2. preventative OR prevention OR preventing OR cardioprotective OR protective OR prophylactic 3. cardiotoxicity OR "Cardiac toxicity" OR "chemotherapy related cardiomyopathy" OR "cancer treatment-related cardiac dysfunction" OR CTRCD OR anthracycline OR doxorubicin OR epirubicin OR Idarubicin OR daunorubicin 4. #1 AND #2 AND #3 5. #4 Limited to exact keyword "Human" **6.** # 5 Limited to Article or Review	3757
PUBMED	"randomized controlled trial " OR "controlled clinical trial" OR randomised OR randomized OR placebo OR randomly OR trial OR rct **AND** preventative OR prevention OR preventing OR cardioprotective OR protective OR prophylactic **AND** cardiotoxicity OR "Cardiac toxicity" OR "chemotherapy related cardiomyopathy" OR "cancer treatment-related cardiac dysfunction" OR CTRCD OR anthracycline OR doxorubicin OR epirubicin OR Idarubicin OR daunorubicin	1461

	Limit to Humans	
COCHRANE	"randomized controlled trial " OR "controlled clinical trial" OR randomised OR randomized OR placebo OR randomly OR trial OR rct **AND** preventative OR prevention OR preventing OR cardioprotective OR protective OR prophylactic **AND** cardiotoxicity OR "Cardiac toxicity" OR "chemotherapy related cardiomyopathy" OR “cancer treatment-related cardiac dysfunction” OR CTRCD OR anthracycline OR doxorubicin OR epirubicin OR Idarubicin OR daunorubicin	415

Table S2. Summary of adverse events amongst included studies.

Trial	Year	Sample size	Symptomatic Heart Failure	Hospitalisation	Death
Massidda[17]	1997	10	NR	NR	NR
Lopez[18]	1998	62	11 (17.7%)	NR	NR
Georgakopoulos[19]	2010	40	3 (7.5%)	NR	NR
Acar[20]	2011	20	NR	NR	NR
Salehi[13]	2011	22	NR	NR	NR
Bosch[21]	2013	37	7 (19%)	NR	8 (24.4%)*
Kaya[22]	2013	18	0	0	0
Liu[23]	2013	20	NR	NR	NR
Elitok[24]	2014	40	0	0	0
Akpek[25]	2015	40	NR	NR	NR
Cadeddu[26]	2016	24	NR	NR	NR
Gulati[27]	2016	30	0	NR	NR
Jhorawat[28]	2016	27	NR	NR	5 (18.5%)#
Pituskin[29]	2016	30	NR	NR	NR
Janbabai[30]	2017	35	NR	NR	NR
Nabati[31]	2017	40	0	NR	0
Avila[7]	2018	96	1 (1%)	NR	2 (2.1%)
Cochera[8]	2018	30	NR	NR	0
Nabati[32]	2019	39	1 (2.6%)	1 (2.6%)	1 (2.6%)

NR – Not Reported, *All of these deaths were described as non-cardiac, secondary to either sepsis or cancer progression, #Cause of death unknown.

Figure S1. Risk of bias for individual studies that used an intention to treat analysis.

Trial ID	Weight	Randomization process	Deviations from intended interventions	Missing outcome data	Measurement of the outcome	Selection of the reported result	Overall Bias
Massida[17]	5.5	Low Risk	High Risk	High Risk	Low Risk	Low Risk	High Risk
Bosch[21]	5.4	Low Risk	High Risk	Low Risk	Low Risk	Low Risk	High Risk
Akpek[25]	5.2	Some Concerns	Some Concerns	Low Risk	High Risk	Low Risk	High Risk
Cadeddu[26]	5.3	Some Concerns	Low Risk	Low Risk	High Risk	High Risk	High Risk
Pituskin[29]	5.5	Low Risk	Some Concerns	Low Risk	Low Risk	Low Risk	Some Concerns
Janbabai[30]	5.2	Some Concerns	Low Risk	Low Risk	Low Risk	Low Risk	Some Concerns
Nabati[31]	5.4	Some Concerns	High Risk	High Risk	Low Risk	Low Risk	High Risk
Avila[7]	5.8	Low Risk	Low Risk	Low Risk	Low Risk	Low Risk	Low Risk
Cochera[8]	5.8	Some Concerns	Some Concerns	Low Risk	High Risk	Low Risk	High Risk

Figure S2. Risk of bias for individual studies that used a pre-protocol analysis.

Trial ID	Weight	Randomization process	Deviations from intended interventions	Missing outcome data	Measurement of the outcome	Selection of the reported result	Overall Bias
Lopez[18]	5.3	High Risk	Some Concerns	Low Risk	High Risk	Low Risk	High Risk
Georgakopoulos[19]	5.2	Some Concerns	High Risk	High Risk	Low Risk	Some Concerns	High Risk
Acar[20]	4.2	Some Concerns	High Risk	High Risk	Low Risk	Low Risk	High Risk
Salehi[13]	5.5	Some Concerns	High Risk	High Risk	High Risk	Some Concerns	High Risk
Kaya[22]	4.9	Some Concerns	High Risk	High Risk	Low Risk	Low Risk	High Risk
Liu[23]	5.3	Some Concerns	High Risk	High Risk	Low Risk	High Risk	High Risk
Elitok[24]	5.4	Low Risk	High Risk	Low Risk	High Risk	Low Risk	High Risk
Gulati[27]	5.5	Low Risk	Some Concerns	Low Risk	Low Risk	Low Risk	Some Concerns
Jhorawat[28]	4.3	High Risk	High Risk	High Risk	Low Risk	Some Concerns	High Risk
Nabati[32]	5.4	Some Concerns	High Risk	Low Risk	Low Risk	Low Risk	High Risk